

CSC1005/CSC1022 CA3 Exam

Functional Dependencies & Normalisation

Instructions:

Write your answers in a single word document and upload (Use Microsoft Word)

Time Limit: 1 hour

QUESTION 1 (Functional Dependencies)

This question uses the **EMPLOYEES** table in the Appendix.

Q 1(a) [6 Marks]

Assuming that a functional dependency is of the form $L \rightarrow R$ where L represents one or more (determinant) attributes and R represents a different set of attributes,

List 4 functional dependencies for the **EMPLOYEES** relation.

Make sure L is different in at least 3 of the 4 functional dependencies.

Q 1(b) [4 Marks]

Consider the relation scheme $R = \{E, F, G, H, I, J, K, L, M, N\}$ and the set of functional dependencies :

$\{E, F\} \rightarrow \{G\}$,

$\{F\} \rightarrow \{I, J\}$,

$\{E, H\} \rightarrow \{K, L\}$,

$\{K\} \rightarrow \{M\}$,

$\{L\} \rightarrow \{N\}$ on R. What is the primary key for R and why?

QUESTION 2 (Normalisation)

This question uses the **EMPLOYEES** and **DEPARTMENTS** tables in the Appendix.

Q 2(a) [5 Marks]

I. In what normal form would you consider the **EMPLOYEES** table to be? Justify your answer with reference to the data and any functional dependencies you can identify.

II. In what normal form would you consider the **DEPARTMENTS** table to be? Justify your answer.

(..... Cotd. on next page)

Q 2(b) [10 Marks]

Taking the **EMPLOYEES** relation, explain the steps you would take to bring this relation to Third Normal Form (3NF).

At each step, clearly identify:

- The specific functional dependency that motivates each decomposition
 - The resulting new relation(s) after decomposition
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Appendix :**Table A: EMPLOYEES Relation**

Emp_ID	Emp_Name	City	Project_ID	Project_Name	Manager_ID	Manager_Name
10011	Sean Kelly	Cork	P201	Cloud Migration	M001	Anne Walsh
10024	Ciara Doyle	Limerick	P201	Cloud Migration	M003	Paul Forde
10037	Roisin Quinn	Dublin	P201	Cloud Migration	M004	Lisa Burke
10055	Brian Farrell	Dublin	P201	Cloud Migration	M004	Lisa Burke
10055	Brian Farrell	Dublin	P202	Data Analytics	M004	Lisa Burke
10068	Sandra Lynch	Cork	P202	Data Analytics	M007	Tom Grady
10071	Deirdre Hogan	Cork	P201	Cloud Migration	M007	Tom Grady
10085	Alan Reilly	Limerick	P202	Data Analytics	M001	Anne Walsh

(Table B : Departments on next page)

Table B: DEPARTMENTS Relation

Dept_ID	Dept_Name	Location	Location_Manager
D01	Software Engineering	Dublin	Ciarán Murphy
D02	Data Science	Cork	Aoife Brennan
D03	Cybersecurity	Galway	Seán Connolly
D04	HR	Dublin	Ciarán Murphy